Dhanavath Babu Mohan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	5.33
EXCEL JUNIOR COLLEGE	Class XII (BIE Telangana)	2023	94.80%
SPR HIGH SCHOOL	Class X (SSC)	2021	100.00%

WORK EXPERIENCE

AI/ML Intern -- Diagonal Matrix

- Developed and enhanced **AI-powered platforms and AI accelerators**.
- Worked with **LLMs, AI agents, and agentic workflows** for intelligent automation.
- Contributed to **API integrations and coding-agent workflows** for enterprise AI solutions.

SCHOLASTIC ACHIEVEMENTS

Secured All India Rank (AIR) 57 (Category Rank) in JEE Advanced 2023.

Secured General Rank 6023 in JEE Advanced 2023, demonstrating strong performance in a highly competitive national examination.

POSITIONS OF RESPONSIBILITY

Class Representative (CR), 1st Year - IIT Hyderabad

Represented the first-year student body at Indian Institute of Technology Hyderabad, acting as a liaison between students and faculty. Facilitated communication regarding academic concerns, schedules, and announcements, ensuring smooth coordination and timely resolution of issues.

Multimedia & PR Co-ordinator, Milan 2024

Contributed to the multimedia and public relations team for Milan 2024, the cultural fest of Indian Institute of Technology Hyderabad. Managed digital content creation, promotions, and social media outreach to enhance event visibility and audience engagement.

Multimedia Co-ordinator, Milan 2025

Led multimedia initiatives for Milan 2025 at Indian Institute of Technology Hyderabad. Oversaw content production, branding, and visual communication strategies to ensure consistent and impactful event representation across platforms.

Operations Co-ordinator, 58th Inter IIT Sports Meet

Served as an Operations Co-ordinator for the Inter IIT Sports Meet. Assisted in managing on-ground operations, logistics, and coordination to ensure the smooth execution of events involving multiple IIT contingents.

SKILLS

Programming Languages: C, C++, JavaScript, Bash, HTML, CSS

Frontend: React, Next.js, Tailwind CSS, Bootstrap, GSAP, Three.js, Framer Motion

Backend: Node.js,Express.js

Database: MySQL, MongoDB

Tools & Platforms: Git, GitHub, Vercel, Netlify, Heroku, Docker, Latex, Bash Scripting, Linux

RELEVANT COURSES

Computer Science: Introduction to Programming, Software Development Fundamentals, Data Structures, Algorithms, Operating Systems, Computer Architecture, Database Management Systems, Theory of Computation, Introduction to Compilers **Mathematics:** Calculus, Discrete Mathematics, Probability and Random Variables, Introduction to Statistics, Linear Algebra, Transformation Techniques

Others: Introduction to Artificial Intelligence, Communication Skills

PROJECTS

RISCV Cache Simulator:

Developed a simulator to emulate RISCV ISA, generate machine code, and simulate cache operations.

Infinite Arithmetic Calculator:

Built a calculator that handles large integer arithmetic (addition, subtraction, multiplication, division) beyond standard data type limits.

Milan Website:

Designed and developed a dynamic website for IIT Hyderabad's annual cultural fest Milan. Integrated a **real-time leaderboard, live score updates**, and a **calendar system** to display event schedules. Implemented **CRUD operations** for admin management of eventsand scores. The website offered a responsive and interactive user experience for both participants and coordinators.

Prayas Website:

Built a clean and informative website for the Prayas student organization, aimed at social outreach. Presented details of initiatives, volunteers, and events. The site emphasized accessibility and ease of updates for non-technical users.

iPhone Clone Website:

Replicated the iPhone user interface using React.js for practice in frontend styling and responsiveness.

Cricket Tournament Management System Website:

Developed a full-stack application as part of a DBMS project to manage registrations, match schedules, and results for cricket tournaments.

JaiTelangana Website:

Developed a comprehensive website showcasing all 33 districts of Telangana. The site includes detailed information about each district, highlights local issues, and celebrates the rich cultural and historical heritage of Telangana.

Memory Management in gemOS:

Implemented custom memory management system calls (mmap, munmap, mprotect) in the gemOS kernel. Extended support to include virtual memory area (VMA) creation, protection handling, merging/splitting VMAs, and page table management. Also handled page faults and laid groundwork for copy-on-write behavior using address space tracking and PFN management.

EXTRACURRICULAR

Cricket NSO Team Member, IIT Hyderabad:

Part of the Cricket NSO team, which identifies and trains talented players in the sport. Engaged in regular practice sessions, skill development, and team coordination under the guidance of experienced coaches to represent the institute in competitive events.